

70. Considerations in Living Kidney Donation · assets 1-9 / 13

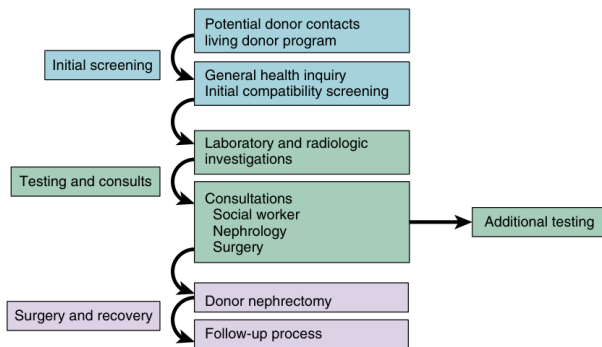


Fig. 70.1 Overview of the living kidney donor evaluation process.

Fig_70_1

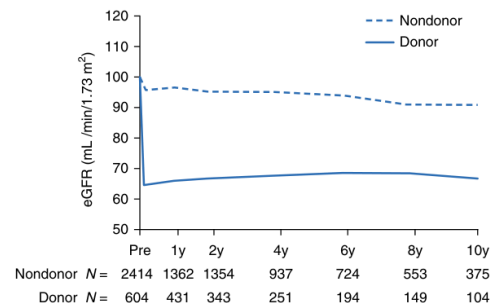


Fig. 70.4 Mean estimated glomerular filtration rate (eGFR) over time in living kidney donors (*solid line*) and matched healthy nondonor controls (*dotted line*). (Reproduced with permission.¹⁹)

Fig 70 4

Guidelines	Year	National/International
Organ Procurement & Transplantation Network (OPTN) Policy 14: Living Donation ⁶⁰	2024	United States
British Transplantation Society (BTS): United Kingdom Guidelines for Living Donor Kidney Transplantation ⁹⁸	2018	United Kingdom
Kidney Disease: Improving Global Outcomes (KDIGO): Evaluation and Care of Living Kidney Donors ¹¹	2017	International
Kidney Paired Donation (KPD) Protocol for Participating Donors ¹⁵	2015	Canada
European Renal Best Practice (ERBP) Guideline on Kidney Donor and Recipient Evaluation and Perioperative Care ¹⁰	2013	Europe
Caring for Australasians with Renal Impairment (CARI) ¹⁰¹	2010 (archived)	Australia/New Zealand

Table_70_1

Medical/Surgical History	Examples	Medical/Surgical History	Examples
Gonorrheal diseases	Hematuria Prostatitis Kidney stones	Infections	Urinary tract infections Hepatitis Human immunodeficiency virus
Cardiac diseases	Coronary artery disease Myocardial infarction Heart failure Arythmia Hypertension	Sexual	Sexually transmitted infections Syphilis Severe sexually transmitted infection Meningitis/Encephalitis Orchitis/Jacobi disease Tuberculosis
Cardiac risk factors	Impaired fasting glucose Impaired glucose tolerance Diabetes mellitus Hyperlipidemia	Cancer	COVID-19 Coronavirus disease 2019 Melanoma Renal cell carcinoma Breast cancer Lung cancer Colon cancer Hemangioma/cancer (benign, lymphoma)
Peripheral vascular disease	Intermittent claudication	Pregnancy-related complications	Monoclonal gammopathy Gestational hypertension Gestational diabetes Preeclampsia/Eclampsia Fetus clubfooting/pain
Pulmonary diseases	Asthma Chronic obstructive pulmonary disease Obstructive sleep apnea Pulmonary edema Inflammatory bowel disease Deep vein thrombosis Pulmonary embolism Blood transfusions	Allergies/medications	Anesthetic reactions Allergies Nephrotoxic Immunosuppressants Chronic pain
Gastrointestinal diseases	Shingles Systemic lupus erythematosus Rheumatoid arthritis		
Hematologic diseases	Drug use/lifestyle Pulmonary embolism Blood transfusions		
Neurologic diseases	Stroke Systemic lupus erythematosus Rheumatoid arthritis		
Autoimmune diseases	Systemic lupus erythematosus Rheumatoid arthritis		
Rheumatologic diseases	Systemic lupus erythematosus Rheumatoid arthritis		
Psychiatric conditions	Anxiety Depression Suicide attempts		

Fig_70_2

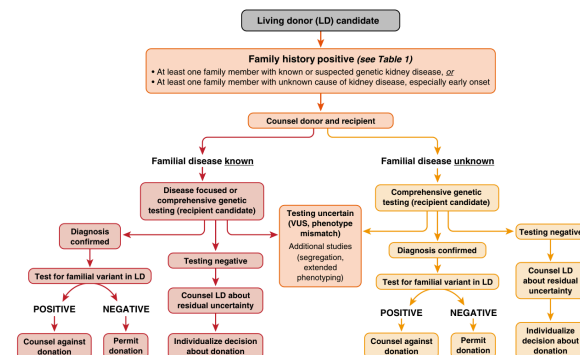


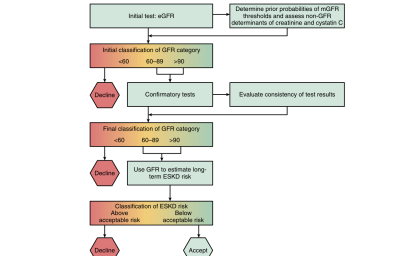
Fig. 70.5 Approach to genetic testing of living donor candidates. LD, living donor; VUS, variant of uncertain significance. (Reproduced with permission.²⁹)

Fig 70 5

Medical/Surgical History	Examples	Medical/Surgical History	Examples
Genitourinary diseases	Hematuria Proteinuria Kidney stones Conjunctiva artery disease Myocardial infarction	Infections	Urinary tract infections Hepatitis Human immunodeficiency virus Syphilis
Cardiac diseases	Heart failure Arrhythmia	Sexually transmitted infections	Severe acute respiratory syndrome Meningitis/Encephalitis Coccidioidi-jakob disease Tuberculosis
Cardiac risk factors	Impaired fasting glucose Impaired glucose tolerance Diabetes mellitus Hypertension Intermittent claudication	Cancer	COVID-19 (Coronavirus disease 2019) Melanoma Renal cell carcinoma Breast cancer Lung cancer Colon cancer Hemolytic cancer (leukemia, lymphoma)
Peripheral vascular disease	Asthma Chronic obstructive pulmonary disease	Pregnancy-related complications	Monoclonal gammopathy Chronic hypertension Gestational diabetes Preeclampsia/Eclampsia Future childbearing plans
Pulmonary diseases	Obstructive sleep apnea Pulpic ulcer disease Inflammatory bowel disease Deep vein thrombosis Pulmonary embolism Bleeding disorders Blood transfusions Strokes	Allergies/Medications	Allergies Nephrotoxic medications Chronic pain medications
Gastrointestinal diseases	Systemic lupus erythematosus Rheumatoid arthritis Gout Anxiety Depression Suicide attempts		Anesthetic reactions Lithium Nonsteroidal antiinflammatory drugs Opioids
Neurologic diseases			
Autoimmune diseases			
Rheumatologic diseases			
Psychiatric conditions			

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Table_70_2_part1

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Fig_70_3

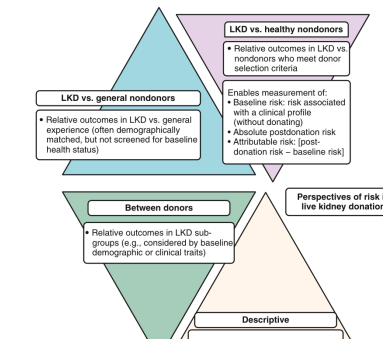


Fig. 70.6 Perspectives of risk in living kidney donation. LKD, Living kidney donor. (From Lentine KL, Seggev DL. Understanding and communicating medical risks for living kidney donors: a matter of perspective. *J Am Soc Nephrol*. 2017;28:12–24.)

Fig 70 6

[illegible]

Table_70_2_part2

Table 70.2 Components of the Living Kidney Donor Evaluation (Continued)				
Medical/Surgical History	Examples	Medical/Surgical History	Examples	
	<i>Trypanosoma cruzi</i> ^a Schistosomiasis ^a Human herpes virus ^a Strongyloides ^a Typhoid ^a Brucellosis ^a West Nile virus ^a Zika virus ^a Chagas ^a Toxoplasmosis ^a COVID-19 (Coronavirus disease 2019) Urine cytology ^b Fecal occult blood test or fecal immunochemical test ^b Prostate-specific antigen ^b Pap smear ^a	Renal	Abdominal ultrasound Abdominal computed tomography or magnetic resonance angiography Measured glomerular filtration rate (serum creatinine) Serum creatinine (eGFR) Cystoscopy ^b Renal biopsy ^b Electrocardiogram 24-hour Holter monitor ^b 24-hour ambulatory blood pressure monitor ^b Echocardiography ^b Stress echocardiography ^b Exercise stress test ^b Coronary angiogram ^b Chest radiograph Pulmonary function test ^b Mammogram ^b	
		Cardiac		
		Pulmonary		
		Cancer screening		
Radiologic Investigations				

Table_70_2_part3

Table 70.3 Contraindications to Living Kidney Donation ^{a, b, c}	
Absolute Contraindications	
Both <18 years old and mentally incapable of making an informed decision High suspicion of donor cancer High suspicion of illegal financial exchange between donor and recipient ABo or HLA human leukocyte antigen) incompatibility without a planned management protocol Impaired kidney function (e.g., glomerular filtration rate <30 mL/min/1.73 m ²) Mannitoluria (eGFR <30 mL/min/1.73 m ²) Hematuria with evidence of kidney disease Diagnosis of IgA nephropathy or Alport syndrome Bilateral/chronic kidney disease Uncontrolled hypertension or hypertension with evidence of end-organ damage Type 1 diabetes mellitus Acute/chronic bacterial or viral infection (Hepatitis B, West Nile virus) Active Mycobacterium tuberculosis or urinary tuberculosis Unresected or incompletely treated syphilis infection Active cancer or incompletely treated cancer History of melanoma, testicular cancer, choriocarcinoma, hematological cancer, monoclonal gammopathy, bronchial cancer, metastatic cancer Treatment with anticoagulation without a planned management protocol Current pregnancy Uncontrolled psychiatric illness Active substance or alcohol abuse disorder	
Relative Contraindications	
Obesity (e.g., body mass index >30 kg/m ² or >30 kg/m ² with other comorbidities) Mannitoluria (eGFR <30 mL/min/1.73 m ²) Two episodes of urinary tract infection (UTI) or renal risk factors Family history of genetic kidney disease with identification of pathogenic variants in the donor candidate Chronic illness (cardiac, pulmonary, gastrointestinal, neurological) Type 1 diabetes mellitus, endocrinopathy, or metabolic syndrome Hemorrhagic disorders with a history of gestational diabetes Prenatal/postnatal women with a history of toxemia in pregnancy Solid-organ transplant Hepatitis C or human immunodeficiency virus without a planned management protocol or clinical trial ^d	

Table_70_3

Table 70.4 Summary of Studies Quantifying Select Long-Term Medical Outcomes in Living Kidney Donors Compared With Healthy Nondonor Controls									
Study, Author, Year ^a , Country	Living Kidney Donors (n)	Healthy Nondonor Controls (n)	Median Donor Age at Donation (years)	Median Donor Age at Follow-up (years)	Median Donor Age at Death (years)	Median Donor Age at Death (years)	Median Donor Age at Death (years)	Median Donor Age at Death (years)	P Value
End-Stage Kidney Failure									
Geiger ¹⁰ (2013), Canada	904	928	7.3	47 (11)	68 (10%)	38 (5.0%)	38 (5.0%)	38 (5.0%)	P = 0.001
Geiger ¹⁰ (2014), Canada	904	928	7.3	47 (11)	68 (10%)	38 (5.0%)	38 (5.0%)	38 (5.0%)	P = 0.001
Acute Kidney Injury Requiring Dialysis									
Geiger ¹⁰ (2013), Canada	904	928	7.3	47 (11)	68 (10%)	38 (5.0%)	38 (5.0%)	38 (5.0%)	P = 0.001
All-Cause Mortality									
Geiger ¹⁰ (2013), Canada	904	928	7.3	47 (11)	68 (10%)	38 (5.0%)	38 (5.0%)	38 (5.0%)	P = 0.001
Geiger ¹⁰ (2014), Canada	904	928	7.3	47 (11)	68 (10%)	38 (5.0%)	38 (5.0%)	38 (5.0%)	P = 0.001
Cardiovascular Mortality									
Geiger ¹⁰ (2013), Canada	904	928	7.3	47 (11)	68 (10%)	38 (5.0%)	38 (5.0%)	38 (5.0%)	P = 0.001
Geiger ¹⁰ (2014), Canada	904	928	7.3	47 (11)	68 (10%)	38 (5.0%)	38 (5.0%)	38 (5.0%)	P = 0.001

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Table_70_4_part1

Table 70.4 Summary of Studies Quantifying Select Long-Term Medical Outcomes in Living Kidney Donors Compared With Healthy Nondonor Controls (Continued)									
Study, Author, Year ^a , Country	Living Kidney Donors (n)	Healthy Nondonor Controls (n)	Median Donor Age at Donation (years)	Median Donor Age at Follow-up (years)	Median Donor Age at Death (years)	Median Donor Age at Death (years)	Median Donor Age at Death (years)	Median Donor Age at Death (years)	P Value
Hypertension									
Geiger ¹⁰ (2013), Canada	904	928	7.3	47 (11)	68 (10%)	38 (5.0%)	38 (5.0%)	38 (5.0%)	P = 0.001
Albuminuria									
Geiger ¹⁰ (2013), Canada	904	928	7.3	47 (11)	68 (10%)	38 (5.0%)	38 (5.0%)	38 (5.0%)	P = 0.001
Kidney Stones With Surgical Intervention									
Thomas ¹¹ (2013), Canada	2,019	20,190	8.4	43 (34-50)	18 (0.79%)	179 (0.88%)	1,044 (5.18%)	1,044 (5.18%)	P = 0.80
Major Gastrointestinal Bleeding									
Thomas ¹¹ (2013), Canada	2,019	20,190	8.4	43 (34-50)	33 (1.6%)	253 (1.25%)	1,044 (5.18%)	1,044 (5.18%)	P = 0.24
Skeletal Fractures									
Geiger ¹⁰ (2013), Canada	904	928	7.3	47 (11)	68 (10%)	38 (5.0%)	38 (5.0%)	38 (5.0%)	P = 0.001
Death									
Geiger ¹⁰ (2013), Canada	904	928	7.3	47 (11)	68 (10%)	38 (5.0%)	38 (5.0%)	38 (5.0%)	P = 0.001
Gestational Hypertension or Preeclampsia									
Geiger ¹⁰ (2013), Canada	904	928	7.3	47 (11)	68 (10%)	38 (5.0%)	38 (5.0%)	38 (5.0%)	P = 0.001

^aData presented as mean (standard deviation) or median (interquartile range).
^bLiving kidney donors were not matched to healthy nondonors on the basis of donor age, sex, or body mass index.

Table_70_4_part2